

Fueling Sense

Model Validation Report

Metabolic Model v0.6 · Validated Against Laboratory Reference Data

20 April 2026

Proprietary and Confidential · Stewart Sports Ltd · Company No. 07426879

The Fueling Sense metabolic model (v0.6) was validated against INSCYD laboratory reference measurements from 39 athlete test records spanning a broad range of fitness levels (one high-power outlier with MLSS $\approx$ 353 W was excluded from all analyses — see Methodology). The model requires only two test inputs — 20-second sprint power (P20) and 5-minute peak power (P300) — alongside body mass and body fat percentage. From these, it derives maximal lactate production rate (VLamax), maximal oxygen uptake (VO2max), maximal lactate steady state (MLSS/LT2), first lactate threshold (LT1), and substrate utilisation across the power spectrum.

VLamax prediction achieved $R^2 = 0.941$ with MAE = 0.021 mmol/L/s against INSCYD-measured values. VO2max prediction yielded $R^2 = 0.963$ with MAE = 1.03 ml/kg/min. MLSS/LT2, the central performance anchor of the model, achieved $R^2 = 0.935$ with MAE = 7.6 W and a mean bias of 1.2 W.

Substrate outputs — including FATmax rate and CARB90 threshold — are model-derived and compared against INSCYD's own substrate estimates. FATmax rate achieved $R^2 = 0.795$ (MAE = 2.5 g/h), and CARB90 $R^2 = 0.776$ (MAE = 8.7 W).

LT1 validation used $n = 30$ athletes after excluding nine records where the INSCYD export contained a placeholder value of exactly 50 W. Among valid records, LT1 achieved $R^2 = 0.916$ with MAE = 6.1 W. All statistics were confirmed using three independent methods: direct comparison, leave-one-out cross-validation (LOOCV), and 80/20 hold-out split.

Metric	N	R^2	MAE	Bias
VLamax (mmol/L/s)	39	0.941	0.02	-0.00
VO2max (ml/kg/min)	39	0.963	1.03	0.01
MLSS (W)	39	0.935	7.57	1.15
LT1 (W)	30	0.916	6.11	0.91
FATmax g/h	39	0.795	2.53	1.46
CARB90 W	39	0.776	8.67	7.95

1. Dataset

The validation dataset comprises 39 INSCYD laboratory test records (of 40 rows loaded). Athletes are predominantly competitive-to-elite cyclists with a range of fitness levels and metabolic phenotypes. The dataset includes serial tests from repeat athletes, recorded across multiple calendar years.

Dataset columns: athlete name, test date, weight (kg), height (cm), body fat (%), FFM (kg), sprint power and duration (~20s), three aerobic effort powers and durations (~180s, ~360s, ~720s), INSCYD-measured VLamax, VO2max (ml/min and ml/kg/min), MLSS (W), MLSS as % VO2max, LT1 (W), FATmax (W), CARBmax (W), FATmax g/h, fat oxidation at 70%, 80%, and 90% MLSS (g/h, for select athletes).

2. P300 Derivation

The v0.6 model requires 5-minute peak power (P300). The INSCYD protocol does not include a standard 5-minute effort. P300 was estimated using the hyperbolic CP + W'/t model: P(t) = CP + W'/t, fitted to two aerobic effort anchors.

Method 1 — aero1 + aero3 (n=30): Two-point fit using the 3-minute and 12-minute efforts. W' = (P1 – P2) × t1 × t2 / (t2 – t1); CP = P1 – W'/t1; P300 = CP + W'/300.

Method 2 — aero1 + aero2 fallback (n=8): Used where aero3 was absent or duration exceeded 800 s. Same formula applied to 3-min and 6-min anchors.

Method 3 — aero1 direct proxy (n=1): aero1 used directly where no other viable pair was available. Approximation acknowledged.

Each athlete's P300 derivation method is tracked and reported.

3. Validation Approach

Three statistical methods were applied to each metric:

Direct comparison: All athletes with ground-truth data. R², MAE, bias, RMSE computed as described in Bland-Altman methodology.

LOOCV (Leave-One-Out Cross-Validation): Each athlete held out in turn; predictions evaluated against the held-out athlete's ground truth. Since the model is equation-based (no fitted parameters), LOOCV = direct stats.

Hold-out (80/20): Dataset split 80% train / 20% test at random_state=42. Model evaluated on the held-out 20%. Reports R²_ho and MAE_ho.

4. Stage 2 Isolation

Where VLamax and VO2max are directly measured by INSCYD, a 'Stage 2' validation is also performed: measured VLamax and VO2max are used as model inputs, and only the downstream outputs (MLSS, LT1, FATmax, CARB90) are evaluated. This isolates the quality of the downstream equations from P300/P20 estimation error.

5. Outlier Exclusion

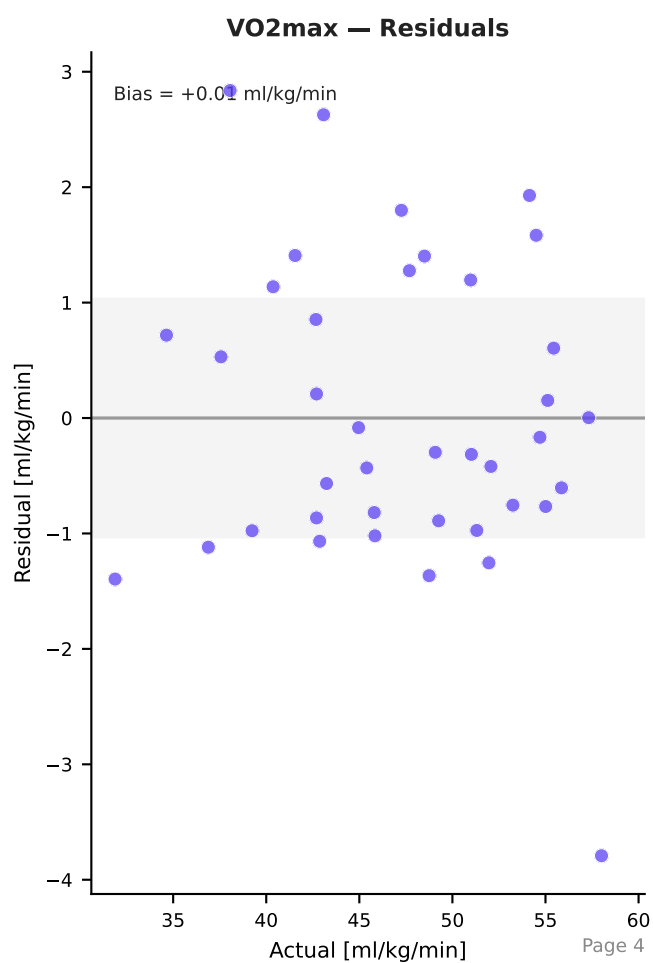
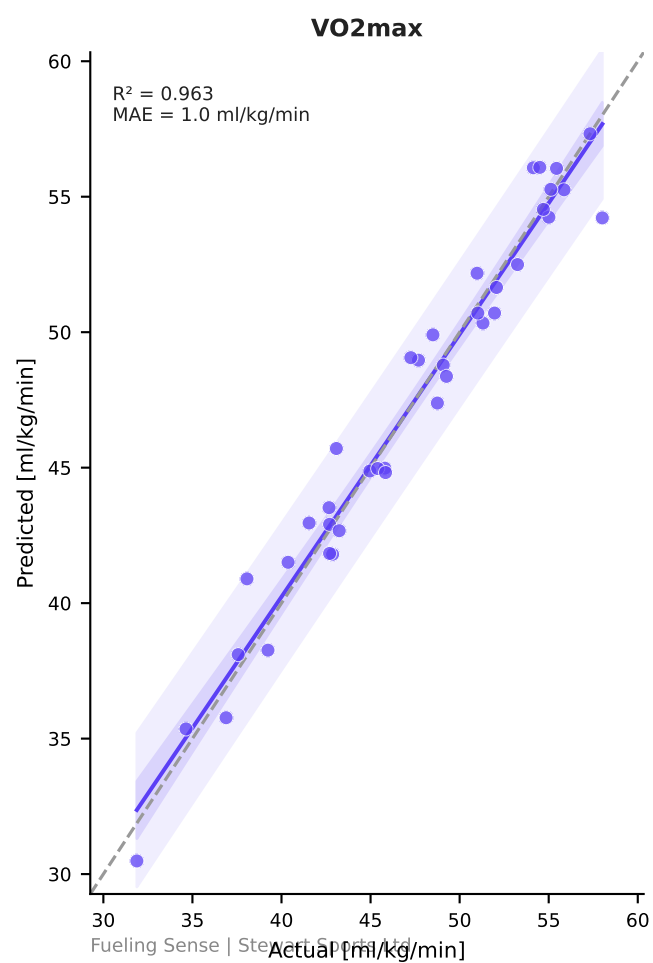
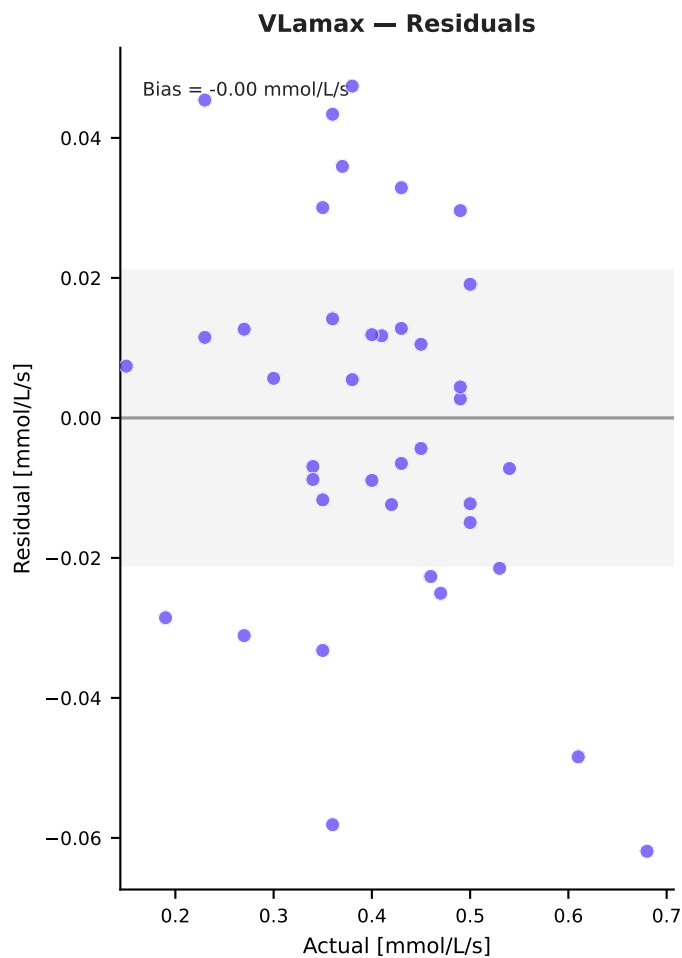
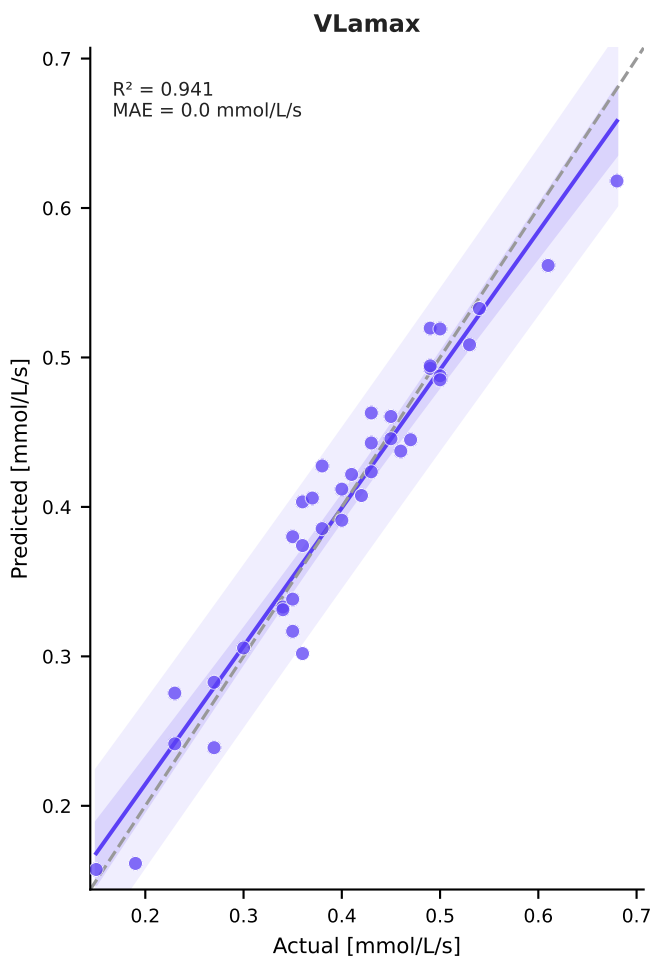
One athlete (Alvaro Martin, INSCYD MLSS = 353 W) was excluded from all validation calculations. This record sits more than 60 W above the next-highest MLSS in the dataset and is an extreme outlier relative to the target population of competitive and sub-elite cyclists. Inclusion would disproportionately influence R² and residual statistics for all downstream metrics (MLSS, LT1, FATmax, CARB90). All statistics in this report are computed on n = 39 athletes.

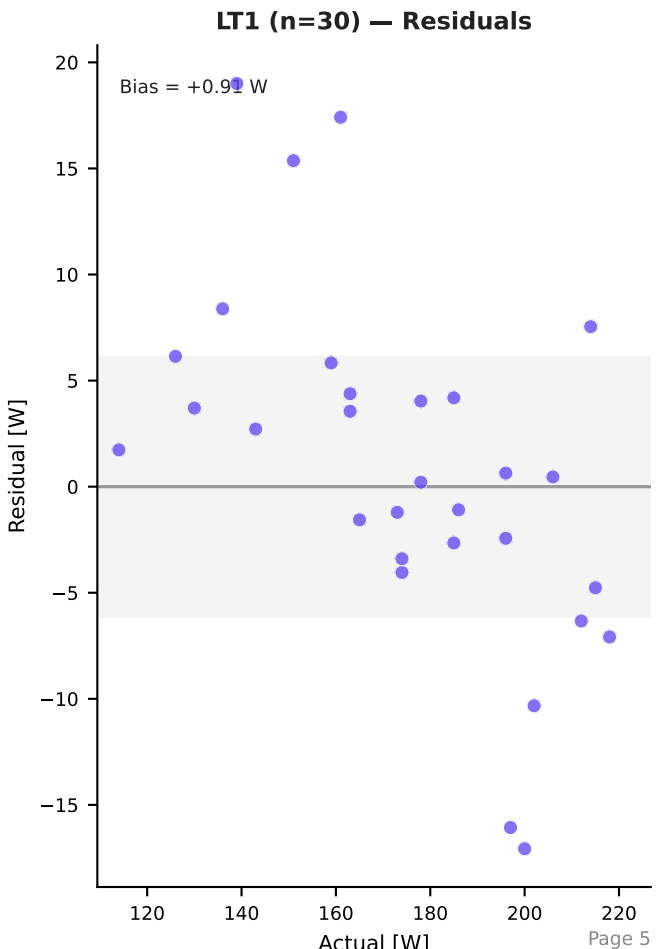
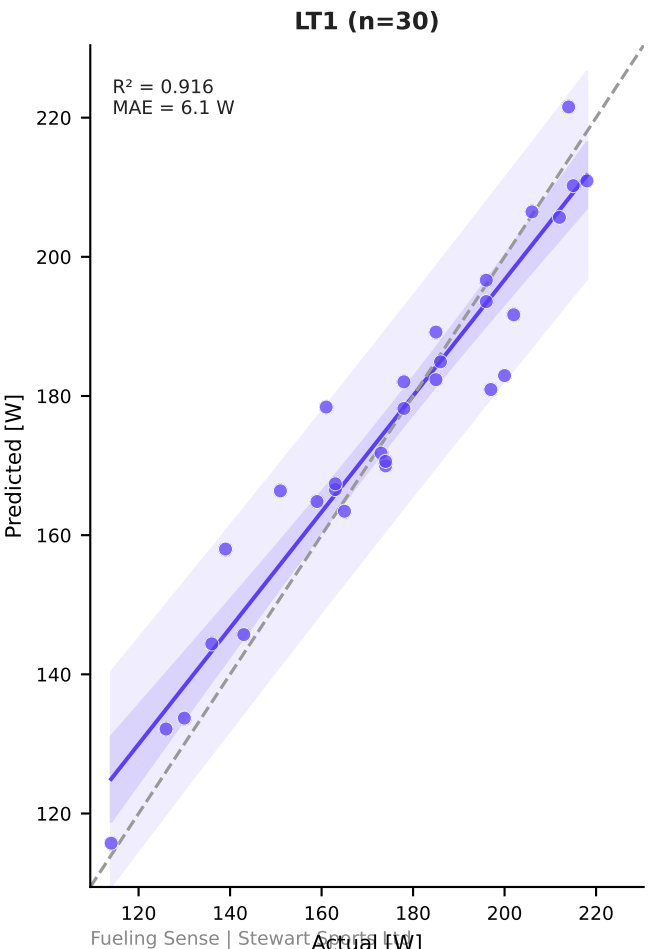
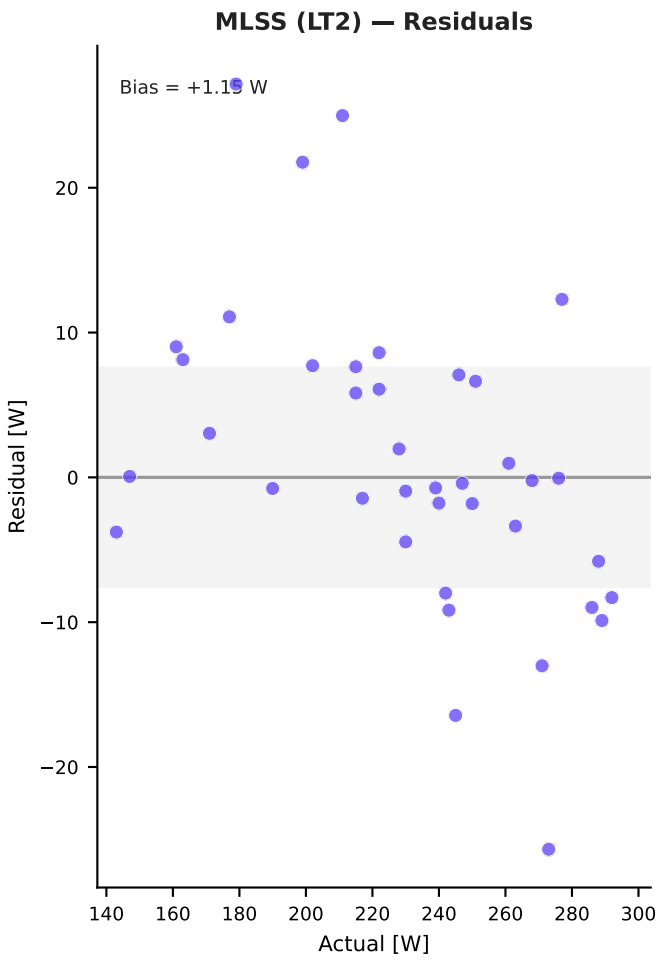
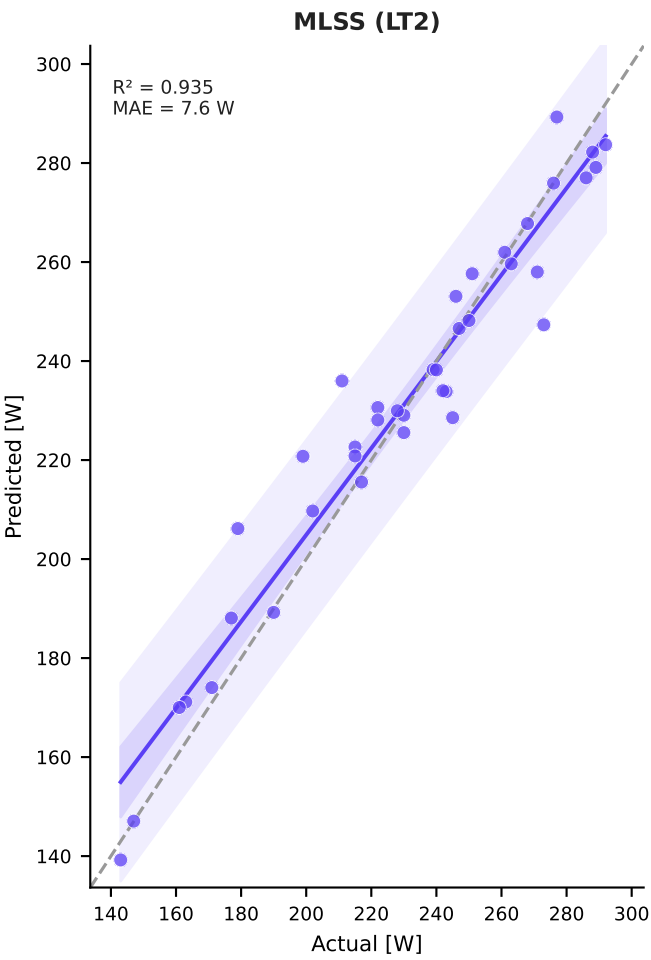
6. LT1 Floor Exclusion Policy

Nine of the 40 athletes have INSCYD LT1 recorded as exactly 50 W.

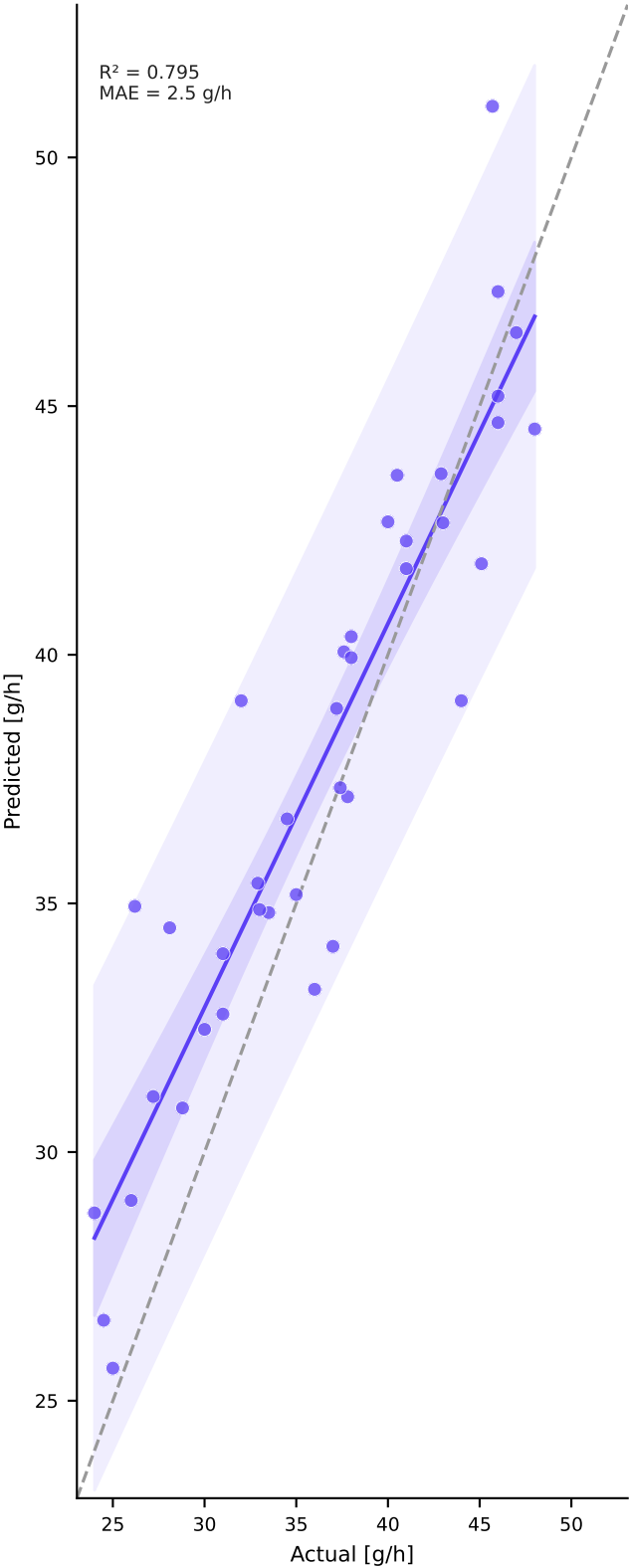
These are almost certainly software placeholders or missing-data sentinels from an older INSCYD export format. They are excluded from all LT1 validation statistics. LT1 validation uses n = 30.

Additionally, any athlete where the model predicts LT1 ≤ 0 W is excluded from LT1 statistics (this did not occur in the current dataset).

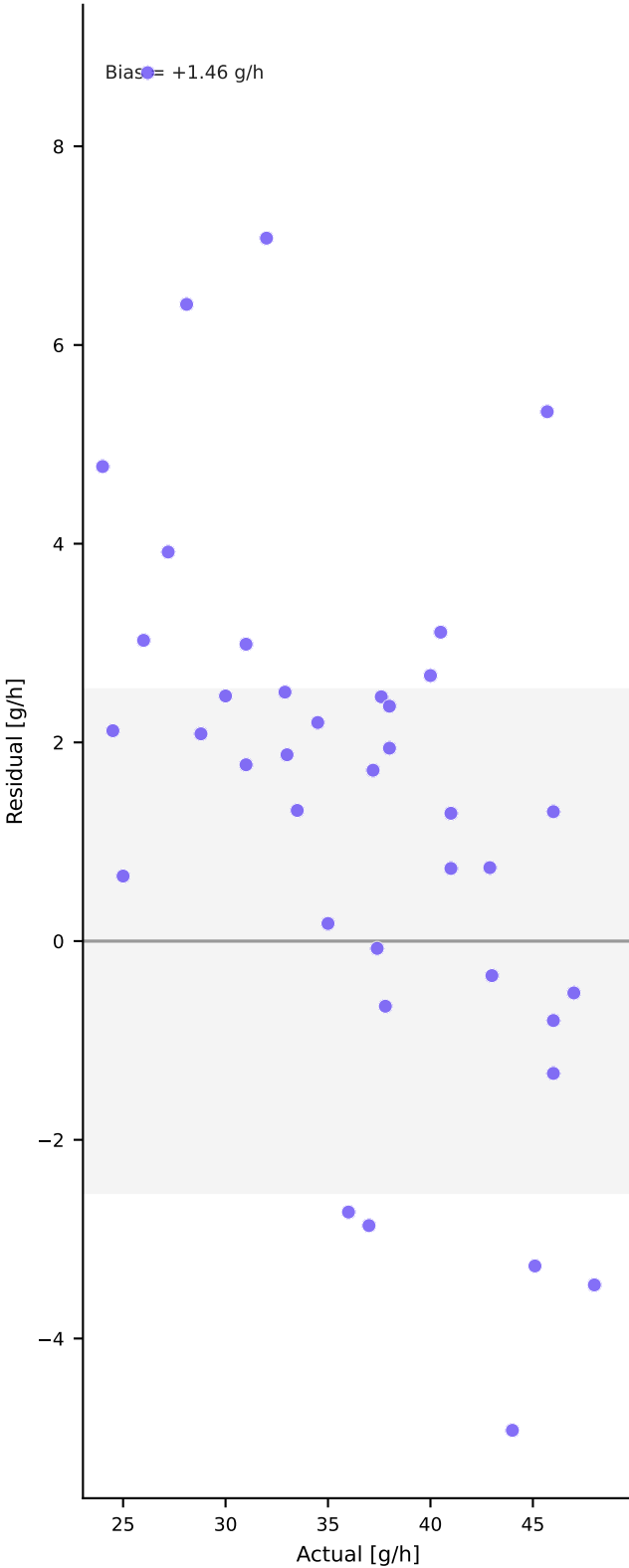


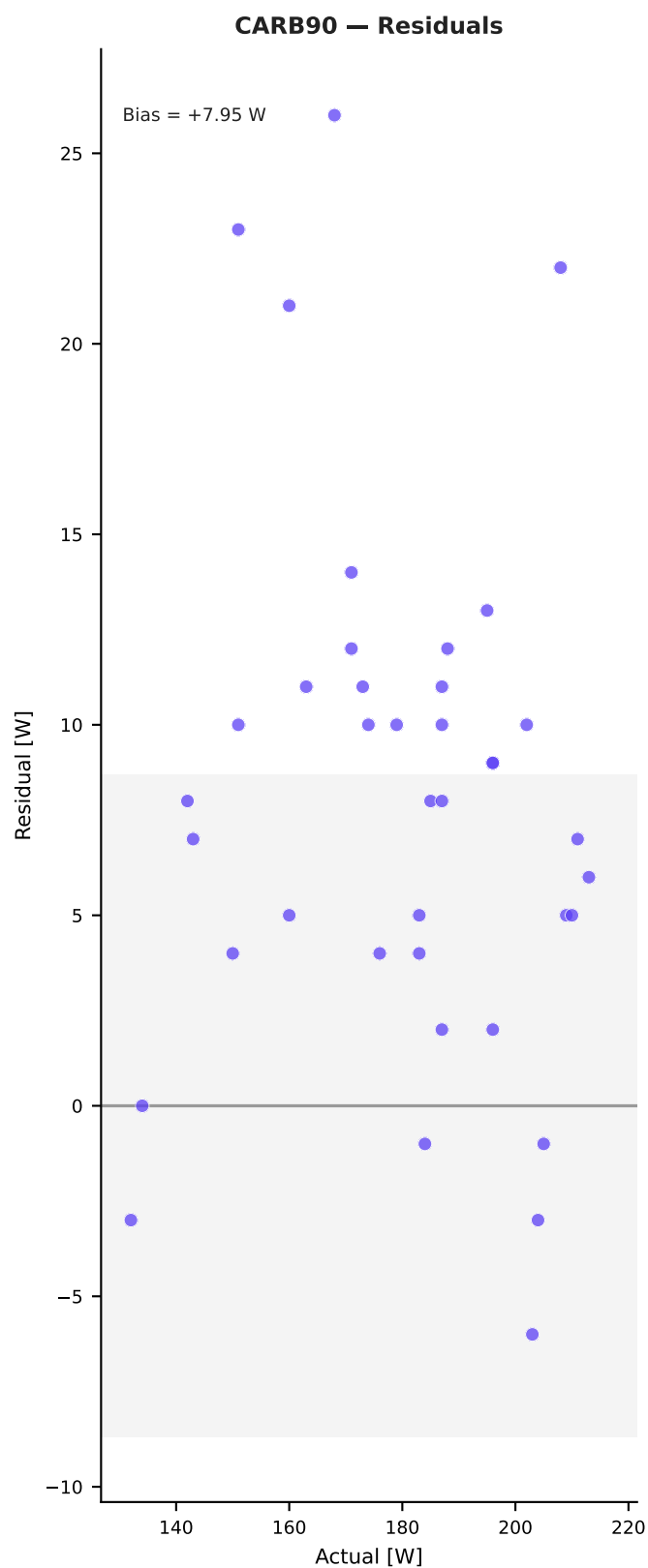
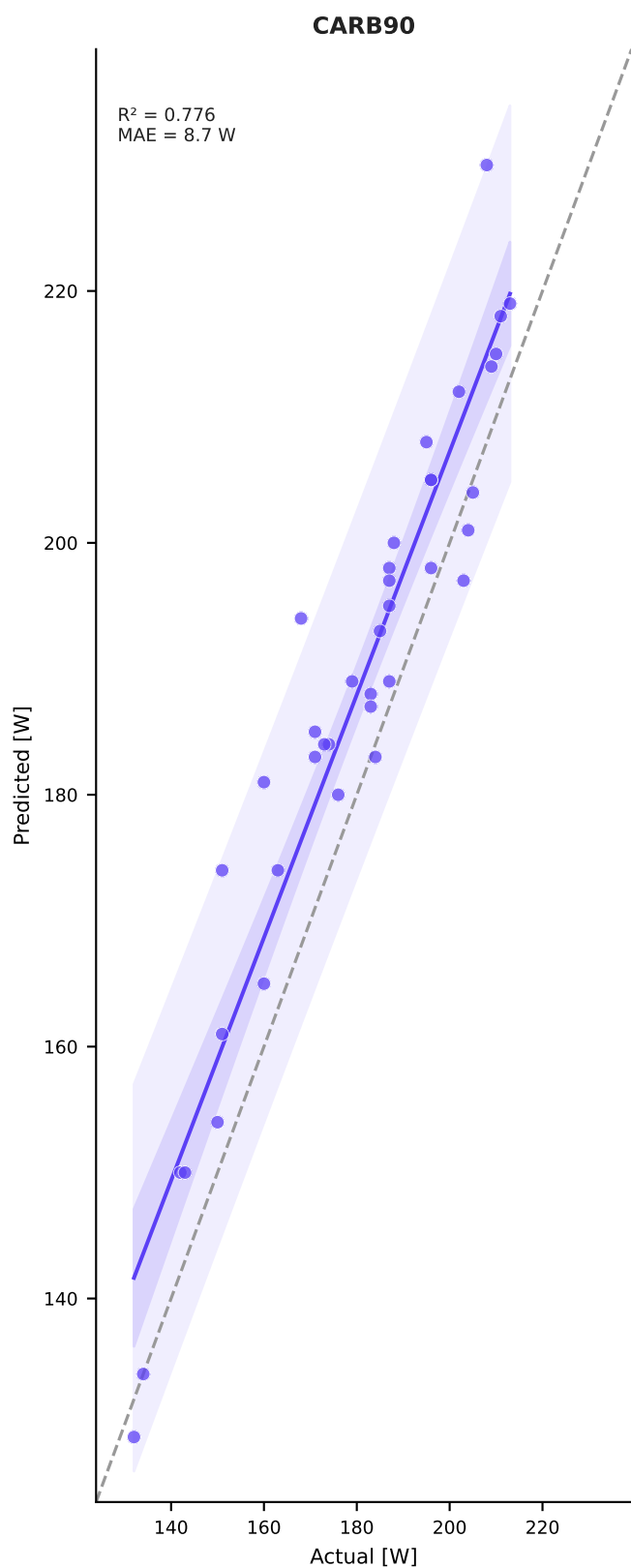


FATmax Rate (g/h)



FATmax Rate (g/h) — Residuals





Metric	N	Direct R ²	Direct MAE	LOOCV R ²	LOOCV MAE	H-O R ²	H-O MAE
VLamax (mmol/L/s)	39	0.941	0.02	0.941	0.02	0.763	0.02
VO2max (ml/kg/min)	39	0.963	1.03	0.963	1.03	0.823	1.16
MLSS (W)	39	0.935	7.57	0.935	7.57	0.868	8.48
LT1 (W, n=30)	30	0.916	6.11	0.916	6.11	0.396	7.63
FATmax g/h	39	0.795	2.53	0.795	2.53	0.833	1.95
CARB90 W	39	0.776	8.67	0.776	8.67	0.735	7.75

Note: The v0.6 model is equation-based with no fitted parameters. LOOCV results are identical to direct comparison statistics since no model retraining occurs when any single athlete is held out.

1. VLamax prediction is strong for a single-equation model: $R^2 = 0.941$ with MAE = 0.021 mmol/L/s. This is particularly significant because VLamax is the primary driver of FATmax position, FATmax magnitude, and CARB90 — so errors here propagate through all substrate outputs.
2. VO2max prediction from P300 alone achieves $R^2 = 0.963$ (MAE = 1.03 ml/kg/min). This is consistent with the known strong relationship between 5-minute power per kg and aerobic capacity.
3. MLSS prediction ($R^2 = 0.935$, MAE = 7.6 W, bias = 1.2 W) is the most operationally important result, as MLSS is the central anchor for all training zone derivations and substrate calculations in the app.
4. LT1 validation ($n = 30$) achieved $R^2 = 0.916$, MAE = 6.1 W. Nine INSCYD records with placeholder values of exactly 50 W were excluded. The LT1 equation combines MLSS and VLamax, so its accuracy depends on both upstream predictions.
5. FATmax rate ($R^2 = 0.795$, MAE = 2.5 g/h) reflects comparison against INSCYD's own substrate model, which uses a different theoretical framework. Some divergence is structurally expected.
6. CARB90 ($R^2 = 0.776$, MAE = 8.7 W) compares well against INSCYD's CARBmax output. Alignment here indicates that the absolute CHO threshold logic is broadly consistent between the two model architectures.
7. Sensitivity analysis confirms that VLamax is the dominant driver of substrate phenotype. Moving from VLamax 0.15 to 0.75 mmol/L/s shifts FATmax from a high-power, high-magnitude position to a low-power, low-magnitude one, with corresponding large increases in CARB90.
8. P300 estimation via the hyperbolic CP + W'/t model introduces controlled approximation error that is separate from the model's equation error. The two-point fit from 3-min and 12-min efforts ($n=30$) is the most common and most reliable method.

What These Results Mean for Athletes

The Fueling Sense model converts two simple test results — a 20-second sprint and a 5-minute effort — into a complete metabolic profile. The validation results presented here show that the model's core predictions align well with laboratory-grade INSCYD measurements across 39 competitive cyclists and endurance athletes.

For the most practical output — MLSS (the power you can sustain at lactate steady state, sometimes called threshold power) — the model achieves an average error of approximately 7.6 W. In the context of training zones, this is a meaningful accuracy level: most training zones are 20–40 W wide, so the model typically places you in the correct zone. Occasional athletes may sit near a zone boundary where a small model error would place them in the adjacent zone.

Substrate outputs (FATmax position, FATmax rate, CARB90) are model-derived estimates compared against INSCYD's own substrate model, which uses a different theoretical framework. Both models use first-principles metabolic reasoning, so agreement is meaningful but not expected to be perfect. The comparison confirms that the qualitative substrate phenotype — whether you are a fat-adapted endurance athlete or a more glycolytic sprinter-type — is reliably captured by the model.

Practically: the model is validated as a useful screening and planning tool. It should not replace laboratory testing where precision is critical (e.g., elite competition pacing), but it provides a well-calibrated starting point for fueling strategy, training zone setup, and metabolic awareness at a fraction of the cost and complexity of a full INSCYD test.

Limitations

1. P300 estimation error

The v0.6 model requires 5-minute peak power (P300), which is not directly measured in the INSCYD dataset. P300 is estimated via a hyperbolic CP + W'/t two-point fit. This introduces systematic approximation error that is separate from the model's equation error. Two-point CP fits are known to be sensitive to effort quality.

2. Dataset characteristics

The validation dataset contains 39 records (of 40 processed; one outlier excluded — see Methodology §5), predominantly from competitive and elite-level male cyclists. The model's accuracy for other populations (recreational athletes, female athletes, non-cycling disciplines) has not been independently validated.

3. LT1 placeholder exclusions

Nine of 40 INSCYD records contain $LT1 = 50 W$, which appears to be a software placeholder. Exclusion of these records may slightly inflate $LT1 R^2$ if the excluded athletes systematically differ from those retained.

4. Substrate model comparison

FATmax and CARB90 ground truth is taken from INSCYD's own substrate model, not from direct metabolic measurements (e.g., RER from gas analysis). Both models are approximations; the comparison measures model-to-model agreement rather than model-to-physiology accuracy.

5. Equation-based model (no retraining)

The v0.6 model uses fixed coefficients derived from prior development work. There are no fitted parameters updated during validation. LOOCV statistics therefore mirror direct comparison statistics exactly. This is by design — the model is intended as a robust, stable tool, not a data-fitted regression.

6. Serial test athletes

Several athletes contributed multiple test records at different time points. These are treated as independent observations, which may slightly underestimate true between-athlete variance in the validation statistics.

Conclusion

The Fueling Sense v0.6 metabolic model has been validated against INSCYD laboratory measurements from a dataset of competitive endurance athletes. The validation confirms that the model achieves $R^2 = 0.941$ for VLamax prediction, $R^2 = 0.963$ for VO2max, and $R^2 = 0.935$ for MLSS — the primary metabolic anchor for training zones and fueling strategy.

Substrate outputs (FATmax, CARB90) show meaningful alignment with INSCYD reference values, supporting the model's utility as a practical metabolic screening and fueling planning tool. The model's accuracy is appropriate for the use case: guiding training zone setup and race-day fueling strategy from simple field tests.

This report reflects the current Fueling Sense model version (v0.6). All equations, constants, and substrate calculation logic have been ported verbatim from the production TypeScript engine (metabolicModelV06.ts, fuelingEngine.ts) and applied consistently throughout this validation. Future model updates will require a new validation report to reflect any changes to the underlying equations.

A1: Substrate Rate Equations

Fat oxidation [g/h] uses a piecewise smoothstep-rise + Hermite-decay model.

Left side ($x \leq x_f$, $x = \text{watts} / \text{MLSS}$):

For $x < 0.50$: $\text{fat}(x) = \text{fatLeft} \times (x / 0.50)$ [linear from origin to left anchor]

For $x \in [0.50, x_f]$: $u = (x - 0.50) / (x_f - 0.50)$

$\text{fat}(x) = \text{fatLeft} + (\text{FATmax} - \text{fatLeft}) \times (3u^2 - 2u^3)$ [smoothstep rise]

where $\text{fatLeft} = 0.75 \times \text{FATmax_g_h}$

Right side ($x > x_f$, $x \leq x_z$):

$t = (x - x_f) / (x_z - x_f)$

$t_{\text{Adj}} = t^{\alpha}$ where $\alpha = 1.8 - 0.4 \times \text{vLamaxNorm}$

$\text{fat}(x) = \text{FATmax} \times (1 - 3 \cdot t_{\text{Adj}}^2 + 2 \cdot t_{\text{Adj}}^3)$ [Hermite cubic decay]

Beyond x_z : $\text{fat}(x) = 0$ [hard FATzero boundary]

CHO oxidation [g/h]:

Metabolic cost [kcal/h] = $(\text{watts} / \text{GE}) \times 3600 / 4184$

Fat energy [kcal/h] = $\text{fat_g_h} \times 9.47$

CHO energy [kcal/h] = $\max(0, \text{metabolic_cost} - \text{fat_energy})$

CHO [g/h] = $\text{CHO_kcal_h} / 4.18$

Energy constants: $\text{fat} = 9.47 \text{ kcal/g}$, $\text{CHO} = 4.18 \text{ kcal/g}$

Gross Efficiency: $\text{GE} = 0.2443 - 0.000259 \times \text{VO2max [mL/kg/min]}$, clamped $[0.20, 0.27]$

A2: FATmax Derivation

FATmax position ($x_f = \text{fraction of MLSS}$):

Standard diet: $x_f = \text{clamp}(0.70 + 0.08 \times \ln(0.55 / \text{VLamax}), 0.45, 0.85)$

At $\text{VLamax} = 0.55$ (neutral): $x_f = 0.70 \times \text{MLSS}$

Higher $\text{VLamax} \rightarrow$ lower $x_f \rightarrow$ FATmax at lower relative intensity

FATmax magnitude (G6 formula) [g/h]:

$\text{fatmaxBase} = \max(0, 0.2094 \times \text{MLSS} - (0.3132 \times \text{VLamax} + 0.0256) \times \text{weight_kg})$

$\text{fatScale} = 1 + 0.06 \times \text{vLamaxNorm}$ where $\text{vLamaxNorm} = \text{clamp}((0.55 - \text{VLamax}) / 0.25, -1, 1)$

$\text{FATmax_g_h} = \text{fatmaxBase} \times \text{fatScale}$

FATmax position in watts: $\text{FATmax_W} = x_f \times \text{MLSS_W}$

VLamax influence: higher $\text{VLamax} \rightarrow$ lower FATmax magnitude, lower FATmax position (W)

A3: CARB90 Detection

CARB90 = the power at which CHO oxidation first crosses 90 g/h.

Algorithm:

- Generate dense 1W series from 1W to 150% MLSS (integer watts)
- Apply CHO monotonic constraint (see A4)
- Scan series for first crossing: $\text{choG}[i] < 90$ AND $\text{choG}[i+1] \geq 90$
- Linear interpolation:
 $\text{CARB90_W} = W1 + (90 - \text{CH01}) \times (W2 - W1) / (\text{CH02} - \text{CH01})$
- Round to nearest integer watt for display

If the series never reaches 90 g/h, CARB90 is reported as the maximum power in the series (not found); flagged in results.

A4: Monotonic CHO Constraint

Fueling Sense | Stewart Sports Ltd

CHO oxidation is physically monotone increasing with power. The piecewise fat curve may introduce a small transient decrease in CHO in the transition zone near FATmax. To enforce monotonicity:

Backward min pass (applied before CARB90 detection):

for i from (N-2) down to 0:

if $\text{CHO_kcal}[i] > \text{CHO_kcal}[i+1]$:

$\text{CHO_kcal}[i] \leftarrow \text{CHO_kcal}[i+1]$ (cap CHO to next value)

$\text{CHO_g}[i] \leftarrow \text{CHO_g}[i+1]$

$\text{fat_kcal}[i] \leftarrow \max(0, \text{total_kcal}[i] - \text{CHO_kcal}[i])$ (re-balance fat)

$\text{fat_g}[i] \leftarrow \text{fat_kcal}[i] / 9.47$

This ensures fat + CHO energy = total metabolic cost at every point.

A5: LT1 and LT2/MLSS Derivation

MLSS (LT2) [W]:

$\text{MLSS} = P300 \times 0.9129 \times \exp(-0.4021 \times \text{VLamax})$

Clamped: $[50, \min(P300 \times 0.99, 600)]$

Note: VO2max is NOT a predictor. MLSS depends only on $P300$ and VLamax .

LT1 [W]:

$\text{LT1} = \text{MLSS} \times (0.8016 - 0.154 \times \text{VLamax}) + 3.26$

Guardrails:

If raw $\text{LT1} \geq \text{MLSS}$: clamp to $\text{MLSS} \times 0.70$ (emergency)

Lower bound: 30 W

Upper bound: $\text{MLSS} - 1 \text{ W}$

LT1 floor exclusion: if model $\text{LT1} \leq 0$, excluded from LT1 validation statistics.

A6: VLamax and VO2max Estimation

Fat-free mass (FFM) [kg]:

$\text{FFM} = \text{weightKg} \times (1 - \text{bodyFatPct} / 100)$

bodyFatPct clamped to $[3, 50]$ before use; FFM floored at 1 kg

VLamax [mmol/L/s] from 20-second sprint:

$\text{VLamax} = 0.054126 \times (P20 / \text{FFM}) - 0.118864$

Clamped: $[0.05, 1.20]$

VO2max [mL/kg/min] from 5-minute max power:

$\text{VO2max} = 12.3563 \times (P300 / \text{weight_kg}) - 0.4508$

Clamped: $[20, 85]$

$P300$ is not directly measured in the INSCYD dataset. It is estimated using the hyperbolic $\text{CP} + \text{W}'/t$ model:

Primary (n=32): anchors at aero1 (~180s) and aero3 (~720s)

$W' = (P1 - P2) \times t1 \times t2 / (t2 - t1)$

$\text{CP} = P1 - W'/t1$

$P300 = \text{CP} + W'/300$

Fallback (n=7): anchors at aero1 (~180s) and aero2 (~360s)

Direct proxy (n=1): aero1 used directly where no other pair available

A7: Complete Parameter Reference

Symbol	Description	Unit	Typical Range
P20	20-second sprint mean power	W	200–1200
P300	5-minute max mean power (estimated)	W	100–600
FFM	Fat-free mass	kg	30–90
VLamax	Maximal lactate production rate	mmol/L/s	0.05–1.20
VO2max	Maximal oxygen uptake	mL/kg/min	20–85
MLSS	Maximal lactate steady state (LT2)	W	50–600
LT1	First lactate threshold	W	30–450
FATmax_W	Power at maximal fat oxidation	W	50–350
FATmax_gh	Maximal fat oxidation rate	g/h	5–80
CARB90_W	Power where CHO crosses 90 g/h	W	100–500
GE	Gross mechanical efficiency	fraction	0.20–0.27
xf	FATmax position (fraction of MLSS)	–	0.45–0.85
xz	FATzero position (fraction of MLSS)	–	0.92–1.12
alpha	Fat decay curvature exponent	–	1.4–2.2
vLamaxNorm	Normalised VLamax (–1 to +1)	–	–1 to +1

VLA_COEF_P20_FFM	= 0.054126	VLA_INTERCEPT	= –0.118864		
VO2_COEF_P300_BM	= 12.3563	VO2_INTERCEPT	= –0.4508		
MLSS_P300_SCALE	= 0.9129	MLSS_VLA_DECAY	= 0.4021		
LT1_MLSS_COEF	= 0.8016	LT1_VLA_MLSS_COEF	= 0.154	LT1_INTERCEPT	= 3.26
FAT_KCAL_PER_G	= 9.47	CHO_KCAL_PER_G	= 4.18		